PROJECT REPORT · DEEP LEARNING SEMESTER PROJECT

Isoline

Reads the ECG.
Knows when not to.



Quality-aware, explainable ECG screening from scans and phone photos, built and evaluated on the PTB-XL dataset.

Author: Talha Salman (560)
Course: Deep Learning · Semester Project
Date: 31 October 2026
Code and weights: https://github.com/talhahSalman/isoline-ecg

Abstract

Electrocardiograms are often shared as printouts, scans or phone photos, not as raw signals. Most deep-learning ECG classifiers need the signal, and most image-based ones answer with the same confidence whether the picture is crisp or crumpled. **Isoline** is a screening pipeline built around one idea: it should check how far its own answer can be trusted. It first rates the quality of the picture, then reads it with an EfficientNet image model, rebuilds the waveform with a classical digitizer and reads that with a second, independent network (a 1D CNN with a Transformer), and finally compares the two. Grad-CAM heat maps and a one-page report explain each result.

We trained and tested on PTB-XL (21,388 ECGs, five diagnostic classes) using the official patient-safe split. On the held-out test set the signal model reaches a macro AUC of **0.929** (0.926 when it only sees what a printout shows), the image model **0.921**, and the quality gate classifies photo quality with **99.1%** accuracy. Answering every picture is right 82.4% of the time; answering only when both models agree is right **89.6%** of the time, on 22.8% of the pictures. The pictures are rendered from PTB-XL signals, not real clinic scans, and Section 8 is candid about what that means.

1 Problem and motivation

The ECG is the most common heart test in the world, and in many clinics, rural health posts and chat conversations the result only exists on paper. Someone prints it, photographs it, and sends the photo. Three things make that hard for software:

- **The signal is gone.** A photo is pixels, not millivolts. A model trained on signals cannot read it directly.
- **Real photos are messy.** Pages are tilted, creased and shadowed; the camera blurs and compresses them. A model that has only seen clean data fails quietly on them.
- **Confidence is not honesty.** A network always produces a number. If nobody checks whether the input was readable, a wrong answer looks the same as a right one.

Isoline treats the third point as the real problem. A screening tool that is sometimes wrong is acceptable if it knows when it might be, and says so. This is the idea behind *selective classification* (Geifman and El-Yaniv, 2017): answer when it is safe, abstain when it is not, and measure the trade-off.

Objectives

- Rate the quality of an ECG picture and refuse the unreadable ones with advice on how to retake them.
- Classify the picture into five diagnostic classes (NORM, MI, STTC, CD, HYP) directly from pixels.
- Rebuild the waveform from the picture and classify it independently, as a cross-check.
- Report how much to trust each answer, and show where the model looked.
- Package everything so a stranger can run it with one command.

2 Dataset

PTB-XL (Wagner et al., 2020) is a public collection of clinical 12-lead ECGs from the PhysioNet archive (Goldberger et al., 2000), released under CC BY 4.0. Each recording lasts 10 seconds and was annotated by cardiologists. We used the 100 Hz version (1,000 samples per lead) of release 1.0.3, which has 21,799 records.

The diagnostic statements are grouped into five *superclasses*. We kept the 21,388 records that carry at least one of them. A record can have several labels, so this is a multi-label problem.

Class	Meaning	Records with this label
NORM	Normal ECG · No abnormality found	9,514
MI	Myocardial infarction · Heart attack, recent or old	5,469
STTC	ST/T-wave change · ST segment or T wave changes	5,235
CD	Conduction disturbance · Bundle branch blocks and similar	4,898
HYP	Hypertrophy · Thickened heart chamber walls	2,649

The five classes. A record can carry more than one.

Split. We used the split recommended by the dataset authors: stratified folds 1 to 8 for training (17,084 records), fold 9 for validation (2,146) and fold 10 for testing (2,158). All thresholds and calibration values were chosen on validation; the test fold was used once for the numbers in this report.

Turning signals into pictures

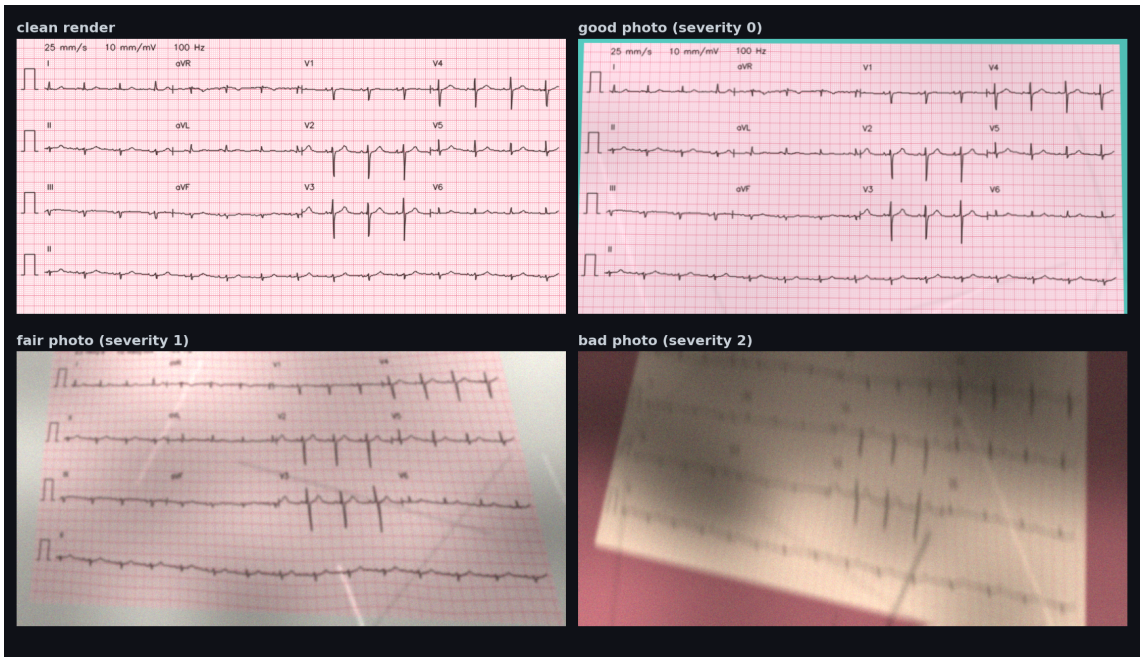
PTB-XL contains signals, not images, so we built the pictures ourselves. A renderer prints every record the way a clinic does: 800 x 400 pixels, 25 mm/s, 10 mm/mV, three pixels per millimetre, the standard 3 x 4 lead layout (each lead shows only 2.5 seconds) plus a 10-second lead II rhythm strip and a calibration pulse on every row. Paper colour (pink, red or grey), ink colour and line thickness vary at random.

A second step, the *camera lab*, photographs each print badly on purpose. It applies rotation, perspective, uneven shadows, creases, blur, sensor noise and JPEG compression at three severities:

Severity	Rotation	Blur	Shadow	Creases	Noise	JPEG quality	Photos
good	up to 1°	0 to 0.6	0 to 0.12	0 to 0.04	0 to 3	85 to 96	8,640
fair	up to 4°	0.9 to 1.6	0.15 to 0.4	0.06 to 0.16	3 to 8	55 to 80	7,379
bad	up to 11°	2 to 3.6	0.4 to 0.7	0.16 to 0.32	8 to 18	25 to 50	5,369

Camera-lab settings. Each record gets one clean print and one photo whose severity is drawn at random.

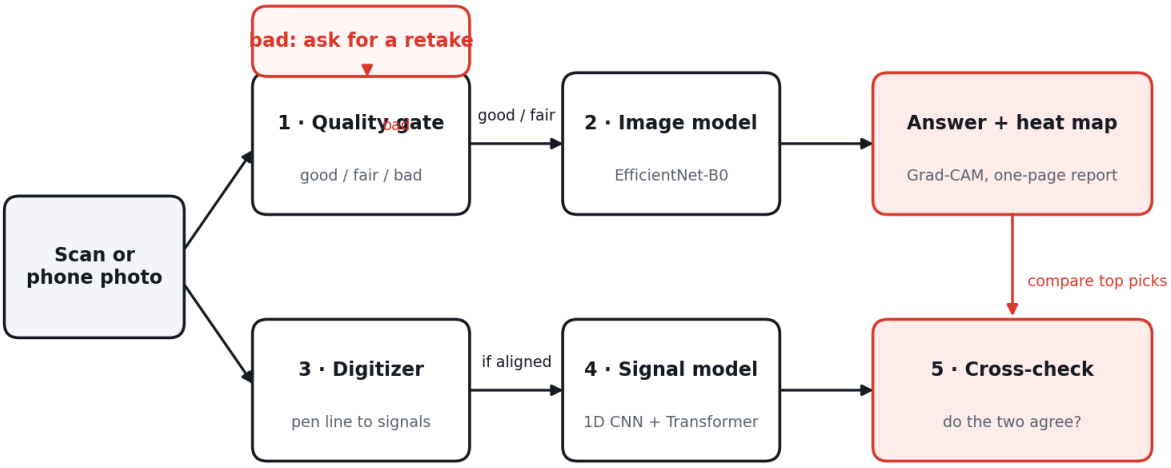
Every record therefore produces two pictures, one clean print and one phone photo, which makes 42,776 images. The severity we applied is the ground truth for the quality gate.



One ECG, shown as a clean print and as phone photos at the three severities.

3 System overview

Isoline is a pipeline of five stages. Every stage can stop the process or change how the answer is described.



The Isoline pipeline. The image model gives the answer; the waveform path is an independent cross-check.

Outcome	When it happens	What the user sees
rejected	The quality gate says the photo is bad	No diagnosis. A request to retake the picture, with tips.
image-only	The photo is readable, but the waveform could not be lined up with the grid	The image model's answer, said to be uncross-checked.
reliable	Aligned, and both models name the same top class	The answer, marked reliable.
flagged	Aligned, but the models disagree	The answer, marked as needing a clinician's look.

The four outcomes. The rules are in isoline/pipeline.py (function decide).

4 Methods

4.1 Image model and quality gate

The image model is an **EfficientNet-B0** (Tan and Le, 2019) pre-trained on ImageNet. The picture is resized to 384 x 192 pixels and normalised with ImageNet statistics. The pooled 1,280-dimensional feature feeds **two heads**: a 5-way head for the diagnoses (independent sigmoids, because labels overlap) and a 3-way head for photo quality (good, fair, bad). Training them together means the features learn to notice both the pattern and the condition of the picture.

The loss is binary cross-entropy for the diagnoses plus half a cross-entropy for the quality. After training we fitted a single **temperature** on the validation set so the probabilities are calibrated (Guo et al., 2017), and chose one decision threshold per class on validation as well.

4.2 Digitizer

The digitizer is classical computer vision, with no learning. It (1) straightens the page to 800 x 400 pixels, (2) turns it into an *ink map* by normalising the illumination and removing the grid colours, (3) follows the pen line column by column inside each lead's band, keeping the run continuous and picking the line's centre, or its far end on steep strokes, and (4) interpolates the result to 100 samples per second in millivolts. Only the part of each lead that a printout actually shows is filled; the rest stays empty.

Tilted photos are the hard part. To avoid trusting a bad rebuild, the digitizer computes an **alignment score**: how much of the ink in the calibration-pulse zone falls exactly where the pulse should be. A clean print scores about 1.0 and photos mostly score below 0.5. Pictures at or above **0.90** (chosen on validation clean prints) go on to the signal model; the others are answered by the image model alone.

4.3 Signal model

IsolineSignalNet reads 12 leads x 1,000 samples. A strided convolution stem is followed by five residual blocks (kernel size 7, widths 64 to 256, each with squeeze-and-excitation, Hu et al., 2018) that shrink the signal to 125 time steps. A 3-layer Transformer encoder (Vaswani et al., 2017) with 4 heads and width 256 and a learned position code lets distant beats talk to each other. Mean and max pooling of the output feed a dropout and a linear layer with five outputs.

It was trained with AdamW (Loshchilov and Hutter, 2019), a one-cycle learning-rate schedule (Smith and Topin, 2019), mixed precision, an exponential moving average of the weights, and light signal augmentation. The best epoch on validation was kept. Thresholds were tuned on validation.

4.4 Making the signal model fit a printout

A printout shows each lead for only 2.5 seconds, except lead II (10 seconds), while the first signal model saw all 10 seconds of every lead. We therefore **fine-tuned it on masked signals** that show exactly the printout layout, with a random gain between 0.85 and 1.1 and small jitter added to imitate digitization error. This version is the one used inside the pipeline.

4.5 Deciding whether to trust the answer

We first tried averaging the two models' probabilities. It did not beat the image model alone, because the rebuilt waveform is noisier than the picture. So the roles are fixed: the **image model gives the answer**, and the signal path is a **witness**. If the witness is available (the picture was aligned) and names the same top class, the answer is reliable; if not, it is flagged.

4.6 Explanation and report

Grad-CAM (Selvaraju et al., 2017) is computed on the last feature map of the EfficientNet for the predicted class and drawn over the picture, so a reader can see which part of the trace drove the decision. The one-page HTML report collects the verdict, the probabilities, the waveforms, the heat map and a plain-language note.

5 Training setup

Everything was trained in Google Colab on a free NVIDIA T4 GPU with PyTorch and torchvision. Pictures were rendered once, saved as JPEG, and streamed from disk by a background thread so the GPU never waited.

	Image model	Layout fine-tune of the signal model
Starting point	ImageNet weights	The full-signal model
Optimizer	AdamW, weight decay 0.01	AdamW, weight decay 0.01
Peak learning rate	0.001	0.0005
Schedule	One-cycle, 20% warm-up	One-cycle, 20% warm-up
Epochs · batch size	10 · 64	12 · 128
Precision · clipping	Mixed (fp16) · gradient norm 1.0	Mixed (fp16) · gradient norm 1.0
Averaging	Weight EMA, decay 0.998	-
Regularisation	Dropout 0.3 · brightness, contrast, noise and shift augmentation	Random gain and jitter on the visible parts
Model selection	Best validation AUC + 0.25 x quality accuracy	Best validation AUC

Training settings for the two models trained in this project's notebook.

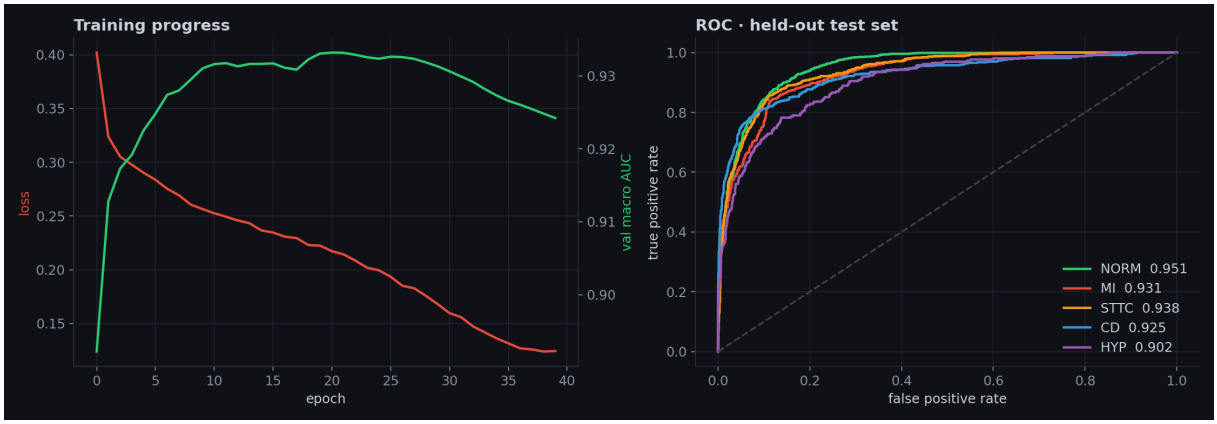
6 Results

All numbers come from the held-out test fold of PTB-XL, ECGs from patients that no model saw during training. AUC is the area under the ROC curve for each class; *macro* means the plain average over the five classes. We report AUC because the classes are unbalanced and a threshold-free score is fair to all of them.

6.1 Per-class discrimination

Class	Signal model (full 10 s)	Signal model (printout layout)	Image model
NORM · Normal ECG	0.951	0.947	0.943
MI · Myocardial infarction	0.931	0.923	0.914
STTC · ST/T-wave change	0.938	0.930	0.934
CD · Conduction disturbance	0.925	0.923	0.910
HYP · Hypertrophy	0.902	0.904	0.905
Macro average	**0.929**	**0.926**	**0.921**

Test AUC. The first two columns use perfect signals, the third uses the pictures (clean prints and photos together).



Signal model: training curve and ROC curves on the test fold.

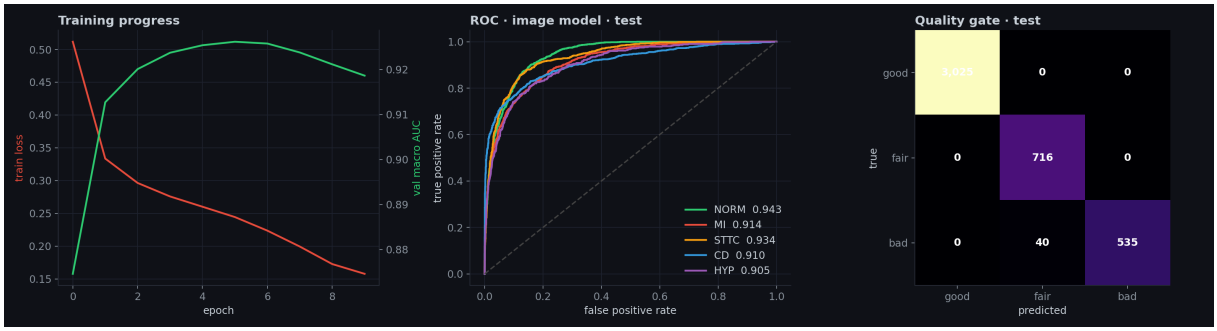


Image model: ROC curves, and the confusion matrix of the photo-quality gate.

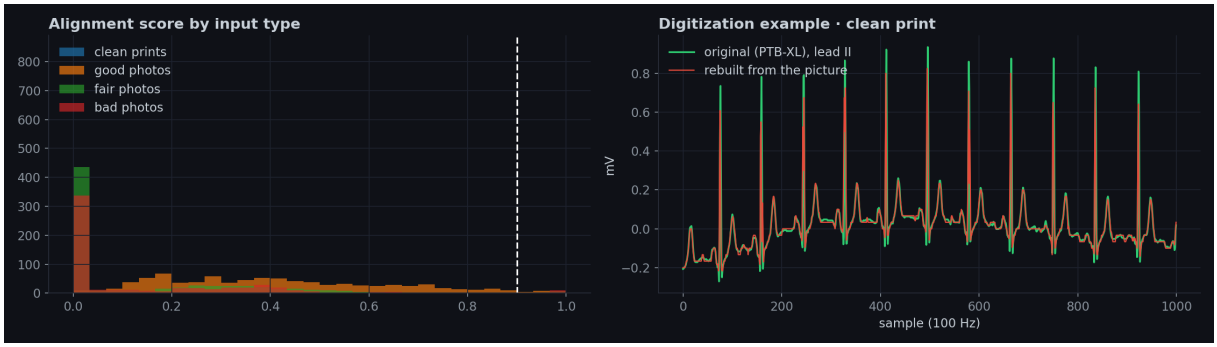
Reading the pictures directly (AUC 0.921) is almost as good as having the perfect signals (0.926 for the same printout layout). Hypertrophy is the hardest class for every model. That is expected: it depends on voltage and axis, which are subtle.

6.2 The quality gate

The photo-quality head labels good, fair and bad photos with **99.1%** accuracy on the test fold. Bad photos are the ones that are refused, so this head is what stops an unreadable picture from getting an answer. Note that the labels are the severities our camera lab applied, so this measures how well the model recognises that kind of damage, not how it judges real phone photos.

6.3 Rebuilding the waveform

On a clean print the digitizer recovers the overall shape of each lead, but a picture carries fewer than one pixel per sample, so sharp R-peaks are slightly clipped. The cost is visible downstream: the signal model scores 0.926 AUC with perfect signals and about 0.814 on rebuilt ones. This is why the signal model is the cross-check and not the main reader. Photos with a strong tilt rarely reach the alignment score of 0.90, so they are answered by the image model alone.



Digitizer: alignment scores of clean prints and photos, and one rebuilt lead.

6.4 Answering less, but more reliably

Policy	Share of pictures answered	Top-1 accuracy
A · answer every image	100.0%	82.4%
B · answer unless the quality gate rejects	87.6%	83.0%
C · B, but withhold when the two models disagree	59.8%	84.9%
D · answer only when both models agree	22.8%	89.6%

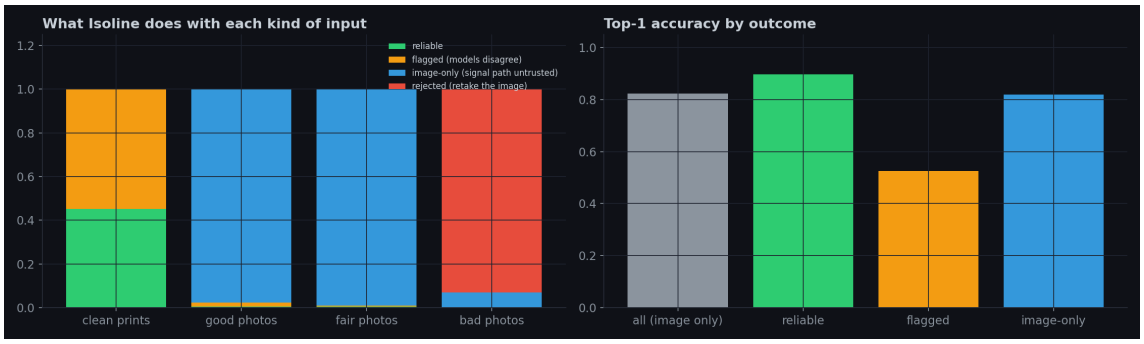
Top-1 accuracy counts a prediction as right if it matches any of the patient's recorded labels. Each row is stricter than the one above it.

The table shows the point of the whole project. If Isoline answers everything, it is right 82.4% of the time. If it answers only when the picture is good enough and both models agree, it is right 89.6% of the time, at the price of answering fewer pictures (22.8%). The user chooses the point on that curve that

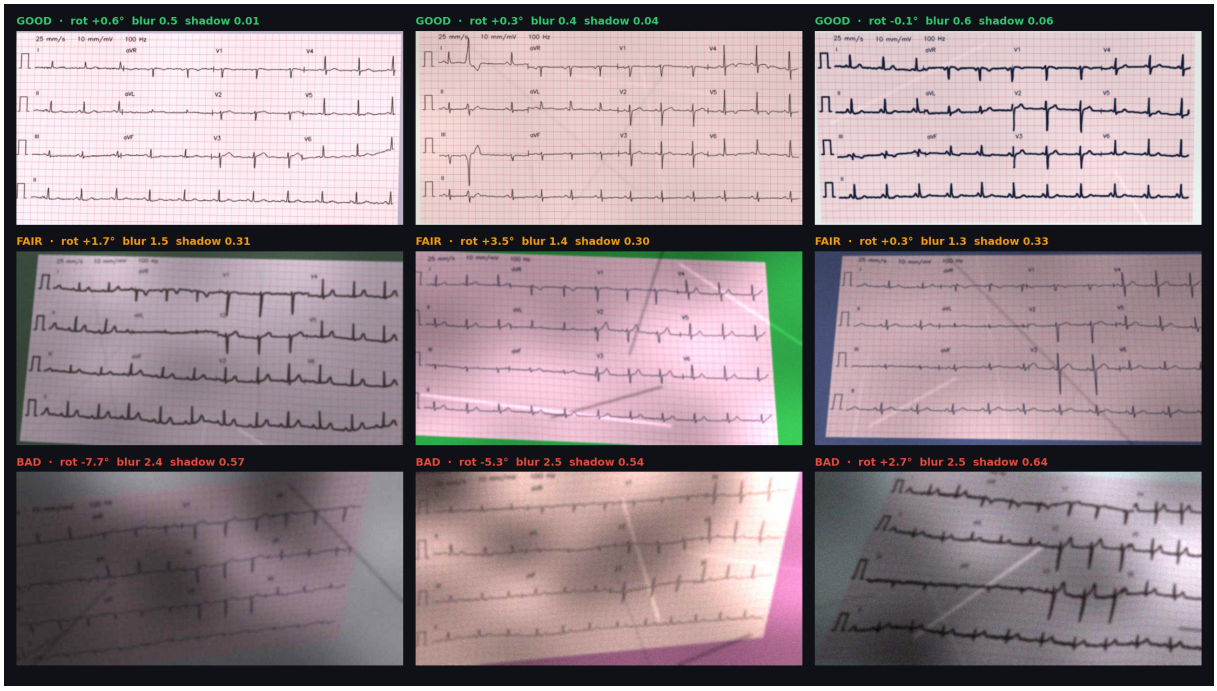
suits the situation.



The trade-off between how many pictures are answered and how often the answer is right.



Reliability analysis of the agreement signal.



Examples of what the image model sees, with its output.

7 Deployment and reproducibility

The project is built so that someone who has never seen it can run it. The repository contains the model code, the trained weights (about 47 MB in total, committed directly), example pictures with their true labels, the training notebook and the website. One command analyses a picture:

```
pip install -r requirements.txt
python predict.py --image samples/ecg_01.jpg --report report.html
```

- **Website.** A static page (no server) on Vercel that shows the dataset, the models, the real results, outputs on test pictures and links to the report, the slides, the video and the code. Example outputs are computed once and stored in the page, so it works even if the live demo is asleep.
- **Live demo.** A Gradio app on Hugging Face Spaces that downloads the code and weights from GitHub when it starts.
- **Checks.** *evaluate.py* runs every example picture through the pipeline and compares with the truth; *predict.py* exits with a clear message instead of a crash on a missing file or weights.

Code, weights and notebook: https://github.com/talhahSalman/isoline-ecg

8 Limitations and ethics

- **Not a medical device.** Isoline is a learning project and a screening aid. It gives no diagnosis and must not replace a clinician, especially for chest pain or breathlessness.
- **Synthetic pictures.** Every picture is a PTB-XL signal printed by our own renderer and degraded by our own camera lab. The models have not been tested on real clinic scans or real phone photos, and real ones will differ. This is the most important limitation, and the main reason to expect lower numbers in the real world.
- **Tilted photos.** Straightening a strongly tilted photo well enough to rebuild the waveform is not solved. Only well-aligned pictures get the cross-check; the rest are answered by the image model alone and say so.
- **Resolution.** With fewer than one pixel per sample, the rebuilt signal loses sharp peaks, which limits the waveform path.
- **Labels.** PTB-XL labels are written by people and are noisy, and five superclasses are a coarse picture of cardiology. The classes are also unbalanced.
- **One dataset.** All data comes from a single collection, so other populations and ECG machines may behave differently.

9 Conclusion and future work

Isoline shows that an ECG reader becomes more useful when it is allowed to say "I am not sure". Reading pictures directly reaches 0.921 macro AUC, a second independent reading through a rebuilt waveform acts as a witness, and asking both to agree lifts accuracy from 82.4% to 89.6% on the cases it answers.

The next steps are clear: (1) test on real scanned and photographed ECGs and fine-tune on them, which matters more than any architecture change; (2) fine-tune the signal model on digitized (not perfect) signals to close the gap between 0.926 and the rebuilt-signal result; (3) a better page finder for tilted photos, so more pictures can be cross-checked; (4) more classes, from the finer PTB-XL sub-classes.

References

[1] Wagner, P. et al. (2020). PTB-XL, a large publicly available electrocardiography dataset. *Scientific Data*, 7, 154.

[2] Goldberger, A. L. et al. (2000). PhysioBank, PhysioToolkit, and PhysioNet. *Circulation*, 101(23), e215-e220.

[3] Tan, M. and Le, Q. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. *ICML*.

[4] Hu, J., Shen, L. and Sun, G. (2018). Squeeze-and-excitation networks. *CVPR*.

[5] Vaswani, A. et al. (2017). Attention is all you need. *NeurIPS*.

[6] Selvaraju, R. R. et al. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. *ICCV*.

[7] Guo, C. et al. (2017). On calibration of modern neural networks. *ICML*.

[8] Geifman, Y. and El-Yaniv, R. (2017). Selective classification for deep neural networks. *NeurIPS*.

[9] Loshchilov, I. and Hutter, F. (2019). Decoupled weight decay regularization. *ICLR*.

[10] Smith, L. N. and Topin, N. (2019). Super-convergence: Very fast training of neural networks using large learning rates. *SPIE*.